QUALITY INFORMATION DOCUMENT

For
OMI_EXTREME_STATE_GLOBAL_trend
Issue: 2.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated, and a row is added to this table. The third column specifies which sections, or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2024/05/15	All	Creation of the document	C. Béchonnet	P. Zunino
2.0	**2025/04/28**	All	Add time series uncertainties. Add a new dataset with global trend maps. Rename product and datasets	C.Béchonnet	P.Zunino

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1) INTRODUCTION

a) Product Name

OMI_EXTREME_STATE_GLOBAL_trend

b) Indicator Evaluation Method

The significant wave height (SWH), expressed in metres, is the average height of the highest one-third of waves. The extreme SWH is defined as the 95th percentile of the daily maximum of SWH over a chosen period and region. The 95th percentile (P95) represents the value below which 95% of the data points fall, indicating higher wave heights than usual.
To compute this OMI, we used the reprocessed L4 SWH product, which merges between one and four missions at a time (WAVE_GLO_PHY_SWH_L4_MY_014_007) and is available from 2002 to 2020.

Then, following Laloue et al. [2024], three regions were selected to compute the regional time series: North Atlantic, South of Australia, Antarctic Circumpolar Current. Both mean and extreme SWH were selected on a seasonal basis to account for the seasonal variability of the waves (January, February and March, JFM, and July, August and September, JAS). Daily data were resampled into seasonal means and P95. Both statistics are in metres. To calculate the 95% confidence interval on the seasonal mean and seasonal extreme time series, a bootstrap method was used by resampling the data and calculating the mean and extreme values 500 times before taking the 97.5 percentile for the upper bound and the 2.5 percentile for the lower bound. Finally, the trends of each time series were then assessed by linear regression as in Timmermans et al. (2020) and are expressed in cm yr-1. The confidence interval for the trends is 90%.

Finally, to produce the global trend maps, daily data were first resampled into seasonal datasets, distinguishing two main seasons of the year (JFM and JAS) to capture the seasonal variability of wave conditions. The mean and 95th percentile of the significant wave height (SWH, in meters) were then computed. Trends were derived using linear regression and are expressed in cm/yr. If the p-value obtained from the linear regression is less than 0.05, the trend is considered significant.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
WAVE_GLO_PHY_SWH_L4_MY_014_007	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-WAV-QUID-014-007.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-WAV-PUM-014-005-006-007.pdf

2) SCIENTIFIC VALIDATION

The scientific validation of this indicator primarily hinges on the quality of the product, it relies on (WAVE_GLO_PHY_SWH_L4_MY_014_007, SEE QUID). A more in-depth validation of this OMI can be found in Laloue et al. (2024).

a) Global trends of SWH 95th percentile and average

The trends of P95 and mean of SWH presented in this OMI (Figure 1) show spatial patterns that have been described in previous studies, particularly Young and Ribal, 2019 and Shimura et al., 2016. As in prior studies, the most pronounced and statistically significant trends occur in the Southern Hemisphere, with the Southern Ocean, the area of south Africa and the Indian Ocean south of Australia emerging as key areas of change. Using satellite altimetry data going from 1985 to 2018, Young and Ribal (2019) identified a wide swath of positive significant trends of 90th percentile of SWH across the Southern Ocean. In this OMI, the patterns of positive and significant SWH trends (both mean and P95) are observable in South of Africa, South Australia and in the South Pacific, coincide mostly with this broad region, but a negative mean SWH trend is detected in the central Indian Ocean around 45°S. Other deviations from earlier findings are also evident. While Young and Ribal (2019), reported significant positive P90 SWH trends in the North Atlantic, our results do not replicate this signal in terms of statistical significance or magnitude and significant trends characterized by complex spatial variations (both increases and decreases in SWH) are identified during the wintertime in the North Atlantic and North Pacific.

Finally, consistent with Young and Ribal (2019), a distinct negative trend of mean SWH is visible in the North Pacific, particularly in boreal winter. This decline in winter SWH during January–March (JFM) in the western North Pacific aligns with projections of decreasing winter wave heights from global climate models (Shimura et al., 2016). Similarly, in agreement with Timmermans et al. (2020), significant positive trends are observable in the North Atlantic and Gulf Stream region in summer (JAS).

Lastly, the results suggest that the significant upper percentile trends (e.g. 95th percentile) are changing at a faster rate than mean SWH trends.

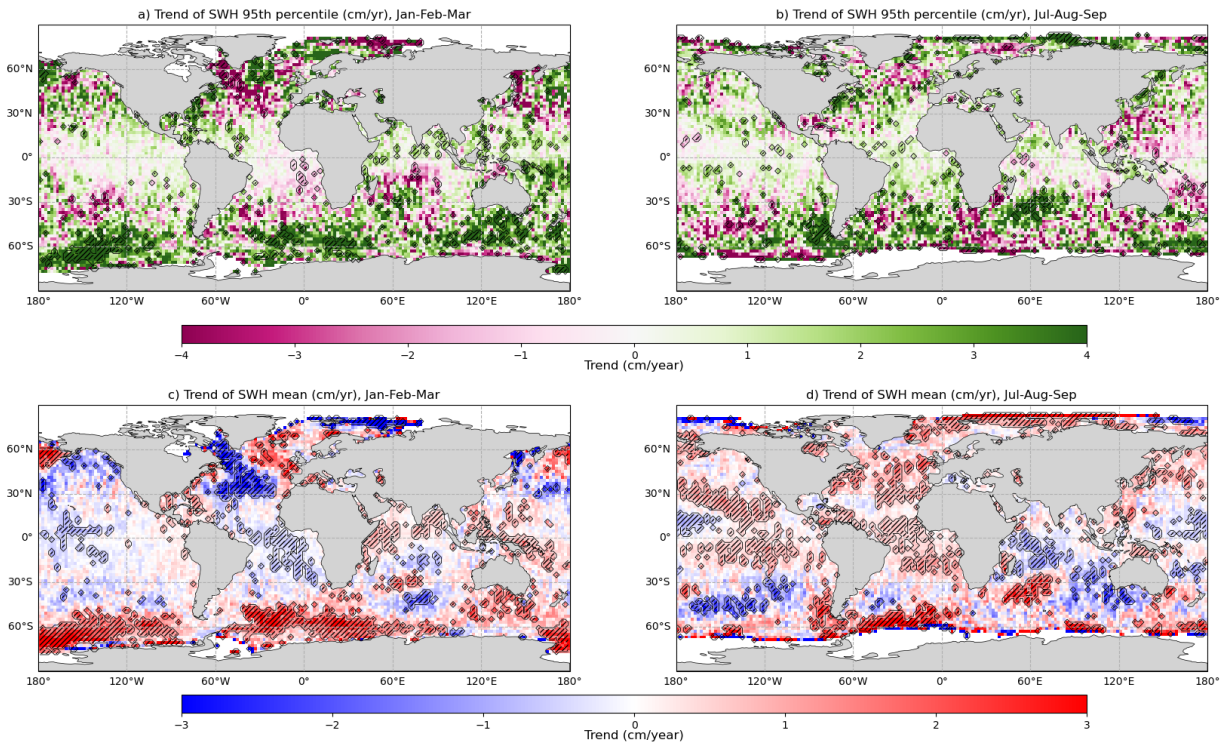

Figure 1 : Global trends of SWH 95th percentile (a and b), and average (c and d). For boreal winter (January, February, March) and boreal summer (July, August, September). Regions with statistical significant trends are striped.

b) Mean and 95th percentile of SWH Time series

As stated in Charles, 2021, one of the key issues when estimating trends in mean and extreme Significant Wave Height (SWH) is the changing number of missions incorporated into the initial product over time. The more missions there are, the better the quality of the initial product. Indeed, with more satellites, there are more along-track measurements available for daily statistics, and the number of observation days also increases. Although the data becomes more accurate over time, by capturing events such as storms more frequently, the SWH distribution is significantly influenced by extreme events.

As explained in Laloue et al. (2024), to assess the impact of the increasing number of satellites on the calculated trends, a new L4 altimetric time series was created by combining data from only two satellites at a time. This series served as a comparison for the L4 multi-mission product used to create this OMI and extended only until 2019, so the comparisons were made only over the period 2002-2019. Figure 2 shows the statistically significant SWH trends for both time-series for the season January-February-March (JFM). After 2008, when more than two satellites were used in the multi-mission product (Figs 2b, 2c, 2d), the mean SWH trends are still nearly identical and not significantly affected by the number of satellites. However, for the P95 daily maximum SWH trends (Figs 2c, 2d) are much more sensitive to the increased number of observations. With the multi-mission product, the trends of P95 seem to be overestimated when compared to the two-satellite product.

Despite this, whether for the mean or the P95, the trend's direction remains consistent across both products and both trends fall within the uncertainty range of the other product. However,

changes in the constellation or the extension of the time series can affect the values of the trends as well as the confidence in them.

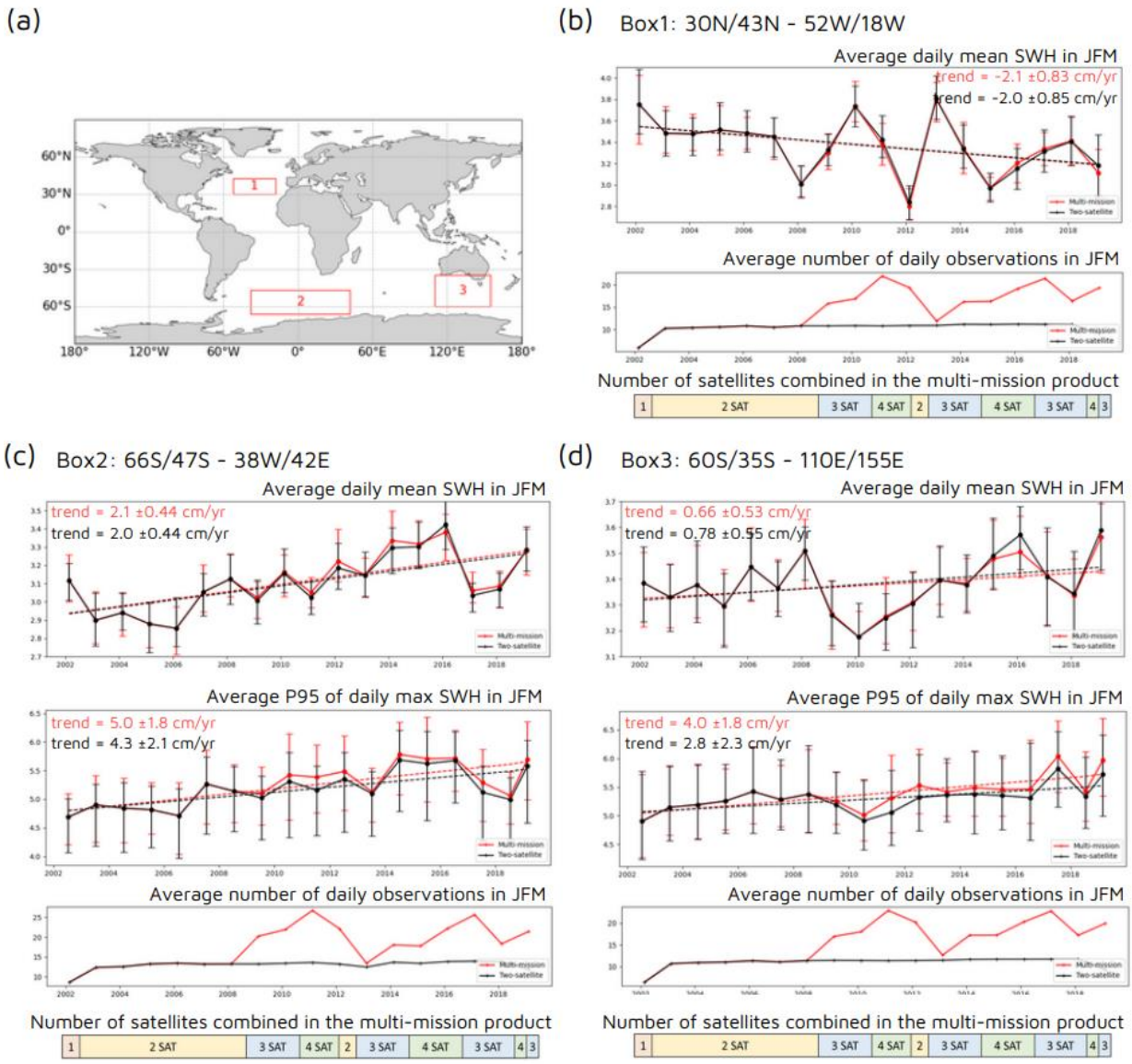

Figure 2: Impact of the number of satellites on the long-term trends in L4 altimetric time series in JFM from Laloue et al. [2024] (a) Location of regional boxes. Box 1: 30° N-43° N, 52° W-18° W; box 2: 66° S-47° S, 38° W-42° E; box 3: 60° S–35° S, 110° E-155° E. (b), (c), (d) seasonal (JFM) mean of daily mean SWH and seasonal P95 of daily maximum SWH time series as well as the seasonal average number of observations. RED: the L4 multi-mission product, BLACK: L4 two-satellite product. Trends are represented by dashed lines and the confidence intervals are represented by error bar. Last subplots is the number of satellites combined in the multi-mission product represented by colored blocks as a function of time as in (Charles, 2021).

3) REFERENCES

Laloue, A., Ghantous, M., Faugère, Y., Dalphinet, A., and Aouf, L.: Statistical analysis of global ocean significant wave heights from satellite altimetry over the past 2 decades, in: 8th edition of the Copernicus Ocean State Report (OSR8), edited by: von Schuckmann, K., Moreira, L., Grégoire, M., Marcos, M., Staneva, J., Brasseur, P., Garric, G., Lionello, P., Karstensen, J., and Neukermans, G., Copernicus Publications, State Planet, 4-osr8, 6, https://doi.org/10.5194/sp-4-osr8-6-2024, 2024.

Charles, E.: EU Copernicus Marine Service Quality Information Document for the Global Ocean L 4 Significant Wave Height From Reprocessed Satellite Measurements Product, WAVE_GLO_PHY_SWH_L4_MY_014_007, Issue 1.0, Mercator Ocean International, https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-WAV-QUID-014-007.pdf, last access: 21 July 2023, 2021

Shimura, T., N. Mori, and M. A. Hemer (2016), Variability and future decreases in winter wave heights in the Western North Pacific, Geophys. Res. Lett., 43, 2716–2722, doi:10.1002/2016GL067924.

Timmermans, B. W., Gommenginger, C. P., Dodet, G., & Bidlot, J.-R. (2020). Global wave height trends and variability from new multimission satellite altimeter products, reanalyses, and wave buoys. Geophysical Research Letters, 47, e2019GL086880. https://doi.org/10.1029/2019GL086880

Young, Ian R., Ribal A., Multiplatform evaluation of global trends in wind speed and wave height.Science364,548-552(2019).DOI:10.1126/science.aav9527